

Tutorial-2
ME-101, (Engineering Mechanics, Semester-II)
Date: 22/01/2024 and Time: 7.55 to 8.50 AM

1. Draw the free-body diagram of the beam as shown in the **Figure 1** that is supported by the pin at B and roller at A. If the intensity of the distributed load acting on the beam is $w = 6 \text{ kN/m}$, determine the reactions at roller A and pin B in **Figure 1**. **[10 marks]**

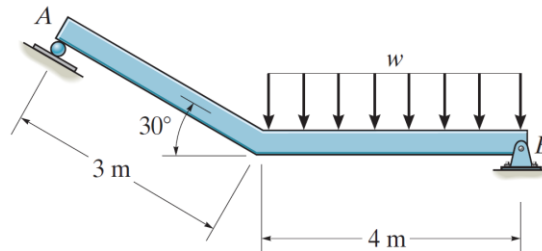


Figure 1

2. Determine whether the truss shown in **Figure 2** is statically determinate or indeterminate (internally as well as externally). Calculate the force carried by each member using the method of joints and show on the figure. Assume all the members are connected using pin joints. **[10 marks]**

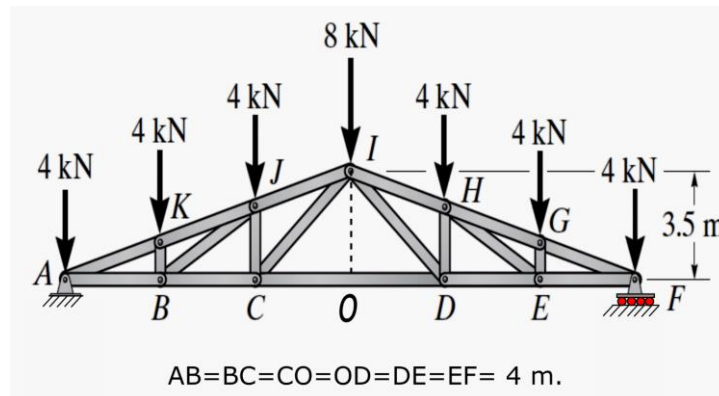


Figure 2

3. For the plane truss shown in **Figure 3**, find out the forces in the members FE , FC , and BC . Consider all the joints to be pin joints. Use method of sections. **[10 marks]**

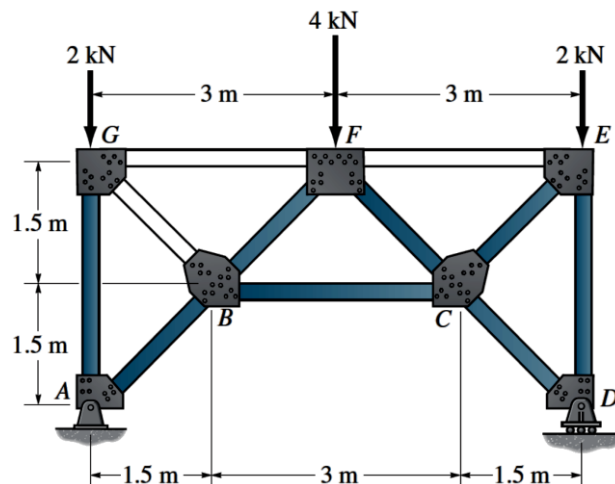


Figure 3

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5. A vertical force P on the foot pedal of the bell crank, shown in **Figure 5**, is required to produce a tension T of 200 N in the vertical control rod. Draw the free-body diagram and determine the corresponding bearing reactions at A and B. **[10 marks]**

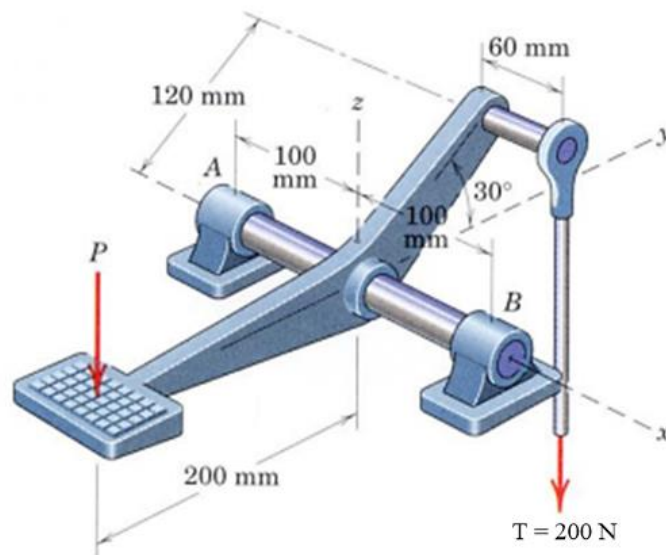


Figure 5